Course Code	**HIGHWAY ENGINEERING**	L	T	P	C
23A01602T		3	0	0	3

Course Objectives:

The objectives of this course are to make the student to:

1. **Understand** the history, importance, and planning aspects of highway developmentin India.
2. **Apply** geometric design principles for highway alignment, sight distance, and curves.
3. **Analyze** traffic characteristics, capacity, level of service, and road safety measures.
4. **Design** flexible and rigid pavements using IRC guidelines.
5. **Evaluate** highway construction materials, testing methods, and maintenance techniques.

Course Outcomes:

Upon successful completion of this course, students will be able to:

1. **Explain** the significance, planning, and alignment of highways.
2. **Design** geometric elements of highways, including curves, gradients, and sight distances.
3. **Analyze** traffic flow, capacity, level of service, and implement road safety measures.
4. **Design** flexible and rigid pavements as per IRC guidelines.
5. **Assess** construction practices, highway materials, and pavement maintenance techniques.

CO – PO Articulation Matrix:-

Course Outcomes	**PO 1**	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PO 12**	**PSO1**	**PSO2**
CO -1	3	2	2	1	-	-	-	-	-	1	-	2	3	2
CO -2	3	3	3	2	2	-	-	-	-	-	-	1	3	3
CO -3	3	3	3	2	2	-	2	1	-	-	-	1	3	3
CO -4	3	3	3	2	2	-	2	1	-	-	-	1	3	3
CO -5	2	2	2	1	3	-	3	1	-	1	1	2	2	2

UNIT – I

PLANNED HIGHWAY DEVELOPMENT in INDIA

Highway development in India – Necessity for Highway Planning- Different Road Development Plans- Classification of Roads- Road Network Patterns – Highway Alignment- Factors affecting Alignment- Engineering Surveys – Drawings and Reports.

UNIT – II

GEOMETRIC DESIGN of HIGHWAYS

Importance of Geometric Design- Design controls and Criteria- Highway Cross Section Elements- Sight Distance Elements- Stopping sight Distance, Overtaking Sight Distance and intermediate Sight Distance- Design of Horizontal Alignment- Design of Super elevation and Extra widening- Design of Transition Curves-DesignofVertical alignment-Gradients- Vertical curves.

UNIT – III

TRAFFIC ENGINEERING STUDIES

Basic Parameters of Traffic-Volume, Speed and Density – Definitions and their inter relation – Highway capacity and level of service concept – factors affecting capacity and level of service - Traffic Volume Studies- Data Collection and Presentation-Speed studies- Data Collection and Presentation- - Road Accidents-Causes and Preventive measures- Accident Data Recording – Condition Diagram and Collision Diagrams.

UNIT – IV		
INTERSECTION DESIGN Conflicts at Intersections- Channelization: Objectives –Traffic Islands and Design criteria- Types of At-Grade Intersections – Types of Grade-Separated Intersections- Rotary Intersection – Concept of Rotary and Design Criteria- Advantages and Disadvantages of Rotary Intersections.		
UNIT – V		
PAVEMENT DESIGN Types of Pavements – Difference Between Flexible and Rigid Pavements – Pavement Components – Sub Grade, Sub Base, Base and Wearing Course – Functions of Pavement Components – Design Factors – Flexible Pavement Design Methods – G.I Method, CBR Method, (As Per IRC 37-2002) –Design of Rigid Pavements – Critical Load Positions - Westergaard'S Stress Equations – Computing Radius of Relative Stiffness and Equivalent Radius of Resisting Section – Stresses in Rigid Pavements – Design of Expansion and Contraction Joints in CC Pavements. Design ofDowel Bars and Tie Bars.		
TEXT BOOKS:		
1. Highway Engineering – S.K.Khanna&C.E.G.Justo, Nemchand& Bros., 9th edition (2011). 2. Transportation Engineering, Volume I, C Venkatramaiah, Universities Press, 2015		
REFRENCE BOOKS:		
1. Principles of Highway Engineering by L.R.Kadiyali, Khanna Publishers 2. Traffic Engineering and Transportation Planning by L.R.Kadiyali andLal- Khanna Publications 9th edition 3. Highway Engineering – Dr. S.K.Sharma, S.Chand Publishers 2014 edition		
Online Learning Resources:		
https://nptel.ac.in/courses/105101087		